

Title of the project: Round-by-Round Win Probability Modeling for Counter-Strike Matches
(Model Comparison + MLP & Transformer Baselines)

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Abstract

This project reframes our analysis from post-hoc variable attribution to round-by-round prediction of the final match winner using team-level cumulative statistics observed up to a given round. Concretely, given Team A's cumulative stats at round r (e.g., kills, deaths, economy/equipment proxies, scoreline), the model outputs the probability that Team A will win the match. We will (1) re-implement and compare common models used in academic esports analytics—logistic regression, random forest, and k-nearest neighbors—and (2) develop two neural approaches: a multi-layer perceptron (MLP) and a Transformer encoder trained on per-round sequences under a strict causal mask (Broms and Nordansjö; Björklund et al.).

Evaluation focuses on accuracy curves by round (e.g., after rounds 3, 5, 10, 15, ...) and probabilistic metrics (log-loss/Brier) across models. For scope and compute balance, we will use ~10,000 CS:GO matches from a Kaggle/HLTV corpus; expanding round-by-round for both teams under CS:GO's MR15 (max 30) yields ~600,000 labeled team-round samples in practice (Machado). We note CS2's MR12 (24 rounds) for future extension ("The Logic Behind Valve's Move to MR12"). Prior work shows supervised models are strong baselines and that neural networks are competitive for Counter-Strike outcome prediction (Broms and Nordansjö; Björklund et al.; Rubin).

Applied impact: outputs support odds-making/gambling (calibrated, round-aware probabilities) and coaching analytics for in-match adjustments and post-match training plans (Rubin).

Background

Esports analytics frequently applies machine learning to forecast match outcomes, with Counter-Strike a natural testbed because of its round-based structure and rich telemetry (Rubin). Academic studies evaluate logistic regression and random forests against baselines such as bookmaker odds or simple heuristics, showing useful—but imperfect—predictive power on hold-out sets (Broms and Nordansjö). Early work also demonstrated that neural networks using team/player representations can be competitive with classical methods in this domain (Björklund et al.). Beyond full-match prediction, researchers have modeled round-level and game-level win probabilities, finding that evolving round statistics can outperform static map-average baselines (Rubin). Practically, calibrated, round-aware probabilities aid odds-making, while coach-facing analyses support tactical timeouts/economy adjustments in-match and post-match training plans focused on patterns that most shifted win probability (Rubin).

Input Data

For the dataset, we will use the CS:GO Professional Matches dataset hosted on Kaggle (HLTV-sourced, 2015–2020). For this study we will sample ~10,000 matches, which—after round-by-round expansion for both teams under CS:GO’s MR15 (max 30) structure—provides ~600,000 labeled team-round examples. While the full dataset has ~25k matches (implying up to ~1.5M samples), we will limit to ~10k for runtime efficiency and reserve the remainder for validation/sensitivity checks (Machado). For CS2’s MR12 (max 24), the analogous cap would be ~480,000 samples (“The Logic Behind Valve’s Move to MR12”).

Features (team-level, cumulative through round r).

- Scoreline through round r (rounds won).
- Combat aggregates: total kills, total deaths, kill–death differential; headshot count/headshot % (if available).
- Economy/equipment proxies: cumulative equipment value spent/held, average equipment value so far (if available).
- Objectives: bomb plants/defuses and related success proxies (if available).
- Survivorship: average surviving players on round wins/losses.
- Optional pre-match aggregates: rating/ranking/Elo-like prior or recent win rate to set prior odds (Björklund et al.; Broms and Nordansjö).

Transformer sequence view: In addition to cumulative snapshots, we will retain the per-round sequence of these team features (and optionally per-round deltas) so the Transformer can learn temporal structure (e.g., momentum, economy swings) under a strict causal mask.

Output Data (Target)

For each team at round r , the model outputs a binary label (team ultimately wins/loses the match) and a probability estimate P (*win / stats through round r*).

Performance Evaluation (The Project’s Real Output)

Primary goal: round-progress accuracy curves. For each model, we compute accuracy at round checkpoints (e.g., after rounds 3, 5, 10, 15, 20, 25, 30) and report log-loss/Brier for calibration. This answers: Given the cumulative team statistics at round r , how accurately can we call the eventual match winner? (Rubin; Broms and Nordansjö; Björklund et al.). We will plot accuracy vs. round number for logistic, random forest, kNN, MLP, and Transformer.

Transformer comparison. For each round r , we compare the Transformer’s prefix-only prediction to the snapshot models. We expect the Transformer to gain earlier (via temporal patterns and economy dynamics) and to match or exceed MLP/Random Forest by mid-game, especially around swing rounds and post-timeout resets (Rubin).

Secondary goals: validation, interpretability, application.

- Sanity checks for leakage: Early plateaus or near-perfect accuracy would indicate leakage; removing pre-match features should lower early-round accuracy if those carried prior signal (Broms and Nordansjö).
- Feature importance/attribution: For Random Forest (and optionally the MLP), we examine feature importances/SHAP; for the Transformer, we inspect attention/gradient-based attributions to highlight decisive rounds (Björklund et al.; Rubin).
- Ablations: isolate contributions by testing pre-match priors alone, score alone, score plus combat/economy, and the complete feature set.
- Applied read-outs:
 - o Gambling/odds-making: Use calibrated round-aware probabilities; evaluate calibration and stability across opponents/maps (Rubin).
 - o Coaching analytics: In-match, trigger tactical timeouts or economy strategy changes when win probability crosses thresholds; post-match, convert attributions into training plans (e.g., early round conversions, retake success, man-advantage conversion) (Rubin).

Success criteria.

- Clear, reproducible accuracy-by-round curves;
- Baselines exceed 50% by early/mid rounds with steady improvement;
- Transformer/MLP match or surpass classical baselines;
- Well-calibrated probabilities and coherent importance patterns that match domain intuition (Broms and Nordansjö; Björklund et al.; Rubin).

Works Cited

Björklund, Arvid, et al. *Predicting the Outcome of CS:GO Games Using Machine Learning*. Bachelor's thesis, Chalmers University of Technology, 2018. Accessed 3 Nov. 2025.

Broms, Erik, and William Nordansjö. *Predicting Counter-Strike Matches Using Machine Learning Models*. Bachelor's thesis in Statistics, Lund University School of Economics and Management, 2024. Accessed 3 Nov. 2025.

Machado, Mateus D. "CS:GO Professional Matches." *Kaggle*, 2019–2020. Accessed 3 Nov. 2025.

Rubin, Adam. "Predicting Round and Game Winners in CSGO." 2022. Accessed 3 Nov. 2025.

"The Logic Behind Valve's Move to MR12." *HDTV.org*, 15 Sept. 2023. Accessed 3 Nov. 2025.